

Things I Don't Understand

A naive audience

Panel 1 — The Thesis: the constructs are cafeteria imports

Every human construct for understanding quantum computing is **internally consistent** — and the imports **contradict each other**. Tools and systems are valid in their own silos; the

Construct	Home silo	What it silently carries
Spiders	Categorical algebra (2D diagrammatic)	Drawing-plane topology; completeness proves facts about the <i>algebra</i> , nothing about spacetime
Pauli webs	Stabilizer formalism / QEC	2+1D code-over-time; in holographic use: entropy, wormholes, black holes
Gadgets	MBQC / compilation	Graph-state resources; rewrite soundness — not physical realization

Fault lines appear exactly where **3+1D particle physics** and **1D thermodynamics** are mixed into a **2D map** without any compatibility check. The diagrams are not wrong — they are where it ends.

Anchor: "ZX Diagrams at the Seam: Spiders, Pauli Webs, Gadgets, and the Cafeteria Problem of Cross-Disciplinary Imports" — 10.5281/zenodo.21992118 (2026-08-18).

Panel 3 — Things I don't understand about QEC

1 · Why the discreteness?

QEC protects a discrete abstraction (gates, qubits, circuits). Is the discreteness physical — or imported from the circuit model? If the abstraction is a choice, fault tolerance is the tax on the choice.

2 · What is a logical qubit, physically?

Hardware is bosonic oscillators with leakage levels; codes assume qubits. The map is 2D; the machine is continuous. Where does the map end?

3 · What does a threshold actually guarantee?

The proofs assume stochastic local noise. Coherent and correlated errors are the seams nobody prices. What is the guarantee outside the model?

4 · Where is the quantum in the loop?

Syndrome → *classical* decoder → feedback. QEC is quantum memory + classical control. Why is it called quantum error correction?

5 · Why isn't the decoder part of the code?

A classical algorithm decides a quantum code's performance. Coding theory and classical computation — two silos, never audited for compatibility.

References (verified 2026-08-18)

ZX Diagrams at the Seam — 10.5281/zenodo.21992118 (concept 10.5281/zenodo.21991895)
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Duncan–Kissinger–Perdrix–van de Wetering — arXiv:1902.03178 · Wan — arXiv:2410.17992

